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Prediction NO₂ Concentration to Mitigation Climate Change Impacts

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INTRODUCTION

Nitrogen dioxide (NO₂) is a key indicator of air pollution, primarily emitted by vehicles and industrial activities. Elevated NO₂ levels contribute to the formation of smog and acid rain, posing significant health risks. Consequently, monitoring and reducing NO₂ concentrations is crucial. This study proposes a machine learning model aimed at predicting NO₂ levels to help mitigate the impacts of climate change.

METHODOLOGY

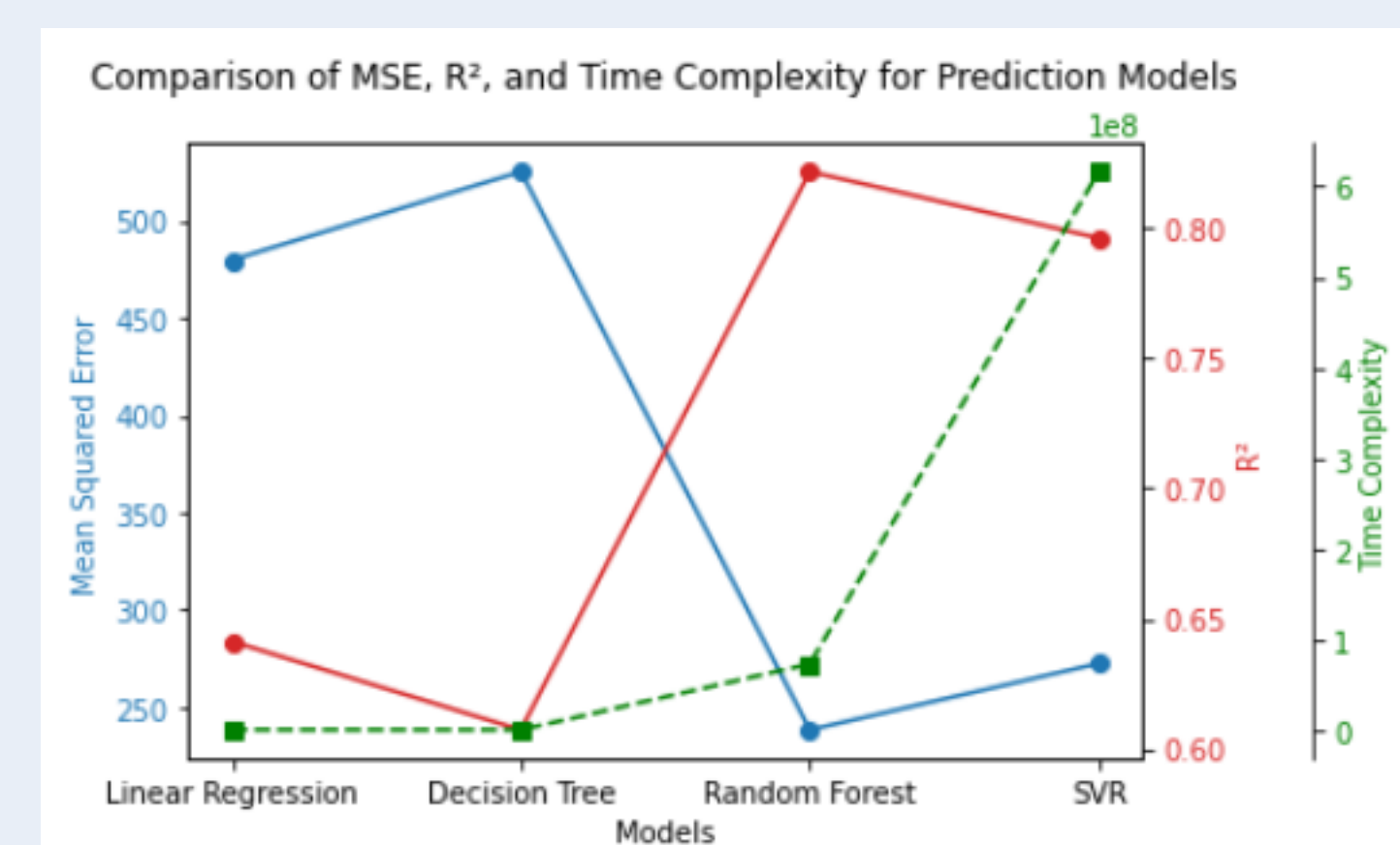
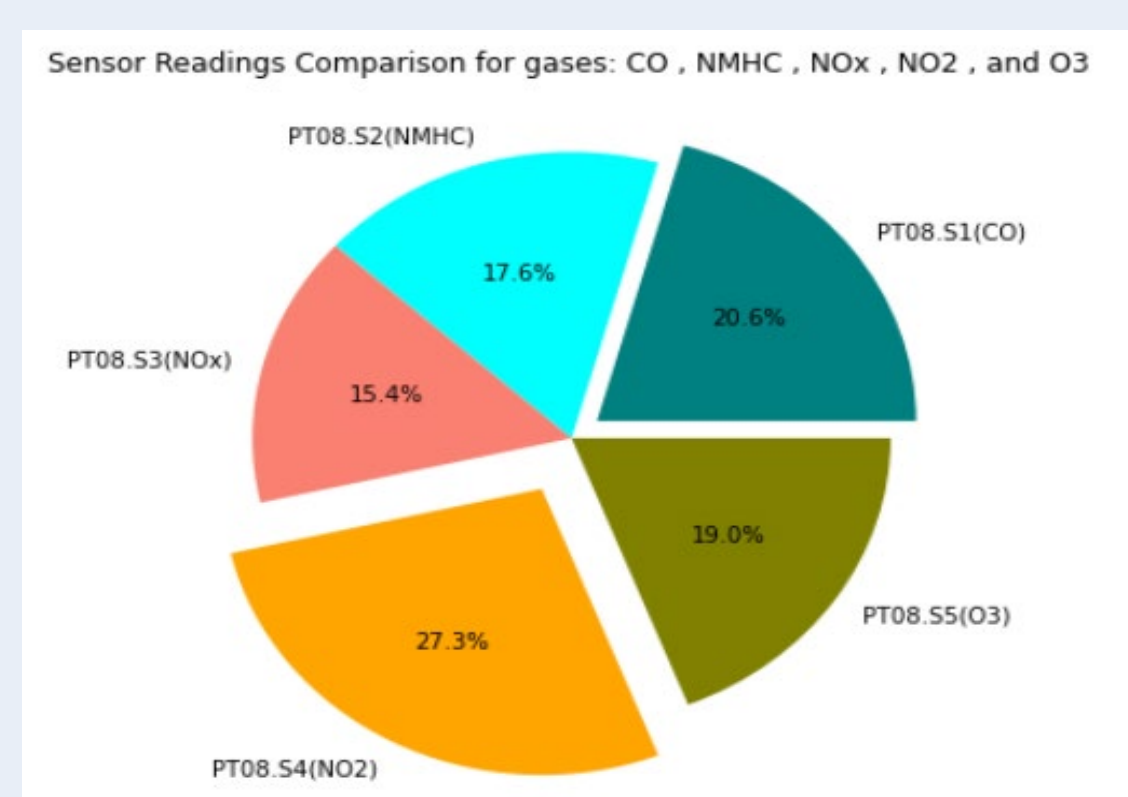
This study compared the performance of four machine models. They are:

1. Linear Regression (LR).
2. Decision Tree (DT) Regression.
3. Random Forest (RF) Regression.
4. Support Vector Regression (SVR).

The four models were evaluated with the Air Quality dataset for one year with 9357 records.

RESULTS & DISCUSSION

- Based on the dataset analysis, NO₂ consistently exhibits the highest concentration among the measured pollutants, constituting 27.3% of total sensor readings. This pronounced prevalence underscores the critical role of NO₂ in environmental pollution relative to other gases. So, NO₂ prioritized as the central focus for the development of targeted mitigation strategies
- The four models evaluated by MSE and R². The results show that the RF model is outperformance compared to other models. The RF has the highest R² (0.82) and the lowest MSE (238.38). In addition, the results show that the RF model strikes a balance between predictive accuracy and computational efficiency, with a reasonable time complexity of O(73448223.62).



CONCLUSION

This study highlights the critical role of machine learning, with the Random Forest model performing best, in predicting NO₂ concentrations. Accurate prediction of this key pollutant can significantly improve air quality management and support climate change mitigation efforts, emphasizing the need for targeted strategies to address NO₂ pollution.

REFERENCES

1. M. Satish, Prakash, S. M. Babu, P. P. Kumar, S. Devi and K. P. Reddy, "Artificial Intelligence (AI) and the Prediction of Climate Change Impacts," 2023 IEEE 5th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA),
2. Athira V, Geetha P, Vinayakumar R, Soman KP (2018) DeepAirNet: applying recurrent networks for air quality prediction. *Procedia Comput Sci* 132:1394–1403. <https://doi.org/10.1016/j.procs.2018.05.068>
3. Abirami S, Chitra P (2021) Regional air quality forecasting using spatiotemporal deep learning. *J Clean Prod* 283:125341. <https://doi.org/10.1016/j.jclepro.2020.125341>
4. Air Quality Dataset, [Air Quality Dataset \(kaggle.com\)](https://www.kaggle.com/datasets/air-quality-dataset).